Collective AI Deliberation: A Multi-Expert Consultation Framework for Human Opinion Refinement

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Abstract

We present a multi-agent AI consultation framework that helps users refine their opinions through structured interaction with domain-specific AI expert clones. Unlike single-LLM chatbots that tend toward sycophancy, our system creates **structured cognitive conflict** by assembling a panel of 5 AI experts with diverse analytical perspectives, each enhanced with domain-specific knowledge profiles (Level 2 AI Cloning). Through a 4-phase consultation pipeline · independent evaluation, interactive dialogue, peer discussion, and final synthesis · the system identifies blind spots in user thinking, challenges cognitive biases, and produces actionable recommendations. We demonstrate the framework with two domain implementations: cryptocurrency investment analysis and sports prediction. Our approach combines techniques from mechanism design, retrieval-augmented generation (RAG), and multi-agent systems to create a consultation experience that is more balanced, rigorous, and informative than interacting with a single AI assistant.

1. Introduction

1.1 The Problem

Large Language Models (LLMs) have become the default tool for seeking advice and analysis. However, single-LLM interactions suffer from several well-documented limitations:

1. **Sycophancy**: LLMs tend to agree with the user's stated position, reinforcing existing biases rather than challenging them (Perez et al., 2023; Sharma et al., 2023).
2. **Perspective Homogeneity**: A single model provides one analytical lens. Complex decisions · investment strategies, career choices, policy judgments · require multiple perspectives that a single system prompt cannot adequately capture.
3. **No Structured Disagreement**: In human expert consultation (e.g., medical second opinions, investment committee decisions), disagreement among experts is a feature, not a bug. It surfaces assumptions, identifies risks, and leads to more calibrated judgments. Single-LLM interactions lack this mechanism.
4. **Shallow Domain Expertise**: General-purpose LLMs lack the specialized analytical frameworks, typical reasoning patterns, and domain-specific heuristics that characterize true domain experts.

1.2 Our Approach

We propose a **Collective AI Deliberation** framework that addresses these limitations through three key innovations:

- **Level 2 AI Cloning**: Each expert agent is enhanced with a domain-specific knowledge profile containing analytical frameworks, few-shot reasoning examples, and characteristic expression patterns. This transforms a generic LLM into a specialized expert clone with consistent analytical behavior.
- **Multi-Phase Consultation Pipeline**: A structured 4-phase process that separates independent evaluation (reducing groupthink), interactive dialogue (enabling user engagement), peer discussion (forcing cross-examination), and final synthesis (producing refined conclusions).
- **Designed Diversity**: Expert panels are constructed with intentionally diverse analytical stances (supportive, skeptical, cautious, neutral) to guarantee that multiple perspectives are represented,

preventing the echo chamber effect.

1.3 Contributions

1. A novel AI expert cloning methodology (Level 2) that uses structured knowledge profiles with few-shot examples to create domain-specialized agents from general-purpose LLMs.
2. A multi-phase consultation pipeline with user-in-the-loop interaction design.
3. Two complete domain implementations (cryptocurrency and sports) demonstrating the framework's applicability.
4. An open-source reference implementation with Streamlit-based user interface.

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2. Related Work

2.1 Multi-Agent LLM Systems

Recent work has explored using multiple LLM agents for improved reasoning. **LLM Debate** (Du et al., 2023) showed that multi-agent debate improves mathematical and factual reasoning. **Society of Mind** (Zhuge et al., 2023) demonstrated benefits of agent role specialization. **ChatEval** (Chan et al., 2023) used multi-agent debate for text evaluation. Our work extends these approaches from factual/reasoning tasks to **opinion refinement**, where the goal is not a single correct answer but a more calibrated, well-considered judgment.

2.2 AI-Assisted Decision Making

Superforecasting research (Tetlock & Gardner, 2015) demonstrated that structured analytical frameworks and diverse perspectives improve prediction accuracy. The **Delphi method** uses iterative expert consultation with controlled feedback. Our system computationally operationalizes these principles with AI agents replacing human experts while preserving the structured disagreement mechanism.

2.3 Prediction Markets and Mechanism Design

Our framework builds on mechanism design principles from prediction markets. **LMSR** (Hanson, 2003) provides a principled way to aggregate beliefs. **Bayesian Truth Serum** (Prelec, 2004) incentivizes honest reporting without requiring ground truth. **Reputation systems** track expert reliability over time. We adapt these mechanisms for the opinion consultation context.

2.4 Retrieval-Augmented Generation

RAG (Lewis et al., 2020) grounds LLM responses in retrieved evidence. **RAFT** (Zhang et al., 2024) trains models to distinguish relevant from irrelevant documents. Our system uses RAG to provide experts with current information while leveraging RAFT-style document evaluation to filter noise.

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3. System Architecture

3.1 Overview

The system consists of four layers:

```
...........................................................
·                    User Interface Layer                    ·
·        Streamlit app with multi-phase interaction         ·
```

```
..........................................................
                         .
..........................................................
.                 Consultation Engine                    .
.    ConsultationPipeline: 4-phase orchestration        .
.    ConsultAgent × 5: knowledge-enhanced expert clones    .
..........................................................
                         .
..........................................................
.               Knowledge & Cloning Layer               .
.    Knowledge Profiles: frameworks + few-shots + style    .
.    Expert Panels: domain-specific expert configurations   .
..........................................................
                         .
..........................................................
.                Information Retrieval Layer              .
.    RAG: YouTube transcripts + Tavily news search        .
.    RAFT-style document mixing and evaluation            .
.    Information partitioning for asymmetry                .
...........................................................
```

3.2 Design Principles

P1: Structured Disagreement Over Consensus. The system is designed to surface disagreement, not suppress it. Expert stances are pre-assigned to ensure multiple viewpoints.

P2: User as Participant, Not Observer. Unlike automated prediction systems, the user actively engages with experts in Phase 2, challenging and being challenged.

P3: Knowledge-Grounded Expertise. Each expert's analytical behavior is anchored in a concrete knowledge profile, not just a persona description.

P4: Parallel Execution for Efficiency. All phases use parallel execution (`ThreadPoolExecutor`) across agents to minimize latency.

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4. Level 2 AI Cloning

4.1 Cloning Hierarchy

We define three levels of AI expert cloning:

| Level | Method | Persona Depth | Cost | |-------|--------|--------------|------| | **Level 1** | System prompt persona (name + description) | Shallow: same LLM, different hat | Low | | **Level 2** | Knowledge profile injection (framework + few-shots + style) | Medium: consistent analytical behavior | Medium | | **Level 3** | Fine-tuned model (LoRA adapter on expert corpus) | Deep: internalized expertise | High |

Level 1 is the approach used by most multi-agent systems · it provides surface-level role differentiation but agents quickly converge to similar reasoning patterns. Level 3 requires substantial training data and compute. **Level 2 is the practical sweet spot**: it provides meaningful behavioral differentiation without the cost of fine-tuning.

4.2 Knowledge Profile Structure

Each expert's knowledge profile is a structured markdown document containing:

```
# Expert Name · Knowledge Profile
```

```
## Archetype
Real-world expert/school of thought this clone is modeled after.
## Core Analytical Framework
### Key Analysis Dimensions
Numbered list of the expert's primary analytical tools and metrics.
### Core Principles/Beliefs
The expert's fundamental assumptions and biases.
## Few-shot Examples
### Example 1: [Scenario Type]
**Question/User Opinion**: [Input]
**Analysis**: [Full expert-style response demonstrating framework
usage]
### Example 2: [Scenario Type]
...
### Example 3: [Scenario Type]
...
## Typical Phrases & Mannerisms
Characteristic expressions that maintain consistency.
```

4.3 Prompt Architecture

The `build_clone_prompt()` function assembles a 4-layer system prompt:

```
···································
· Layer 1: Identity               ·  Expert name, role, expertise areas
···································
· Layer 2: Knowledge Profile      ·  ~80% of total prompt
·   - Analytical framework        ·  Framework, examples, mannerisms
·   - 3 few-shot examples         ·  loaded from markdown files
·   - Typical phrases             ·
···································
· Layer 3: Behavioral Rules       ·  Stay in character, use framework,
·                                 ·  be honest, give concrete advice
···································
· Layer 4: RAG Context            ·  Retrieved documents for current
topic
···································
```

The knowledge profile constitutes approximately 80% of the system prompt, making the expert's analytical DNA the dominant signal in the LLM's context window.

4.4 Crypto Domain Expert Clones

| Expert | Archetype | Stance | Key Framework | |--------|-----------|--------|---------------| | ······ | Willy Woo / Glassnode | Neutral | MVRV, NUPL, exchange flows, on-chain metrics | | ······ | Raoul Pal / Lyn Alden | Skeptical | Global liquidity cycles, real rates, debt/GDP | | ······ | Arthur Hayes / Messari | Supportive | Narrative cycles, ecosystem metrics, halving | | ······ | Nassim Taleb / Kelly Criterion | Cautious | Max drawdown, VaR, position sizing, volatility | ······ | a16z Policy / Coin Center | Neutral | SEC actions, global regulatory framework, compliance |

4.5 Sports Domain Expert Clones

| Expert | Archetype | Stance | Key Framework | |--------|-----------|--------|---------------| | ······ | FiveThirtyEight / Nate Silver | Neutral | ELO, Net Rating, Monte Carlo simulation, r² || ····· | Zach Lowe / Zonal Marking | Neutral | Matchup analysis, coaching adjustments, system fit | | ······ | Sports medicine perspective | Cautious | Injury recurrence rates, load

management, age curves | | ······· | Pinnacle / Sharp market theory | Neutral | CLV, implied probability, sharp vs public money | | ······· | Bill Simmons / narrative school | Supportive | Historical analogies, championship DNA, intangibles |

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5. Multi-Phase Consultation Pipeline

5.1 Pipeline Overview

```
User Input: topic + opinion + domain
         .
         .
    ···········     RAG retrieval: YouTube transcripts
    · Context  ····· + news articles, RAFT-style mixing
    · Retrieval·     + information partitioning
    ···········
         .
         .
    ···········     5 experts evaluate independently
    · Phase 1 ····· Parallel execution (ThreadPoolExecutor)
    · Evaluate·     Output: assessment, agree/challenge points,
    ···········     key insight, confidence score
         .
         .
    ···········     User reads evaluations
    · Phase 2 ····· User chats 1-on-1 with experts
    · Interact·     Conversation history maintained
    ···········     User advances when satisfied
         .
         .
    ···········     Each expert sees all others' evaluations
    · Phase 3 ····· Experts respond, update, identify blind spots
    · Discuss ·     Parallel execution
    ···········
         .
         .
    ···········     Each expert gives final assessment
    · Phase 4 ····· Recommendations + risk warnings
    · Finalize·     Confidence tracking (Phase 1 · Phase 4)
    ···········
         .
         .
    ···········     Aggregate insights, blind spots,
    · Report  ····· recommendations, risk warnings
    ·Synthesis·     Confidence shift visualization
    ···········
```

5.2 Phase 1: Independent Evaluation

Each expert independently evaluates the user's opinion without seeing other experts' views. This prevents anchoring and groupthink.

Input: Topic, user opinion, RAG context (expert-specific via information partitioning)

Output structure:

- `ASSESSMENT`: Overall evaluation (support / partial support / oppose)
- `AGREE_POINTS`: What the expert agrees with and why
- `CHALLENGE_POINTS`: What the expert challenges with evidence
- `KEY_INSIGHT`: Most important thing the user should know
- `CONFIDENCE`: 0-100% confidence in the assessment

Design rationale: By requiring structured agreement AND disagreement, we prevent the sycophancy failure mode. The expert must find something to challenge even if they largely agree.

5.3 Phase 2: Interactive Dialogue

The user reads all Phase 1 evaluations and can engage in free-form conversation with any expert. This is the **user-in-the-loop** component that distinguishes our system from fully automated multi-agent debates.

Key features:

- Per-expert conversation history is maintained
- Expert responses are context-aware (topic + original opinion + conversation history)
- No forced structure · the user drives the conversation depth

Design rationale: Users often have follow-up questions or want to challenge specific expert claims. This phase allows the user to test the robustness of expert reasoning.

5.4 Phase 3: Expert Discussion

Experts review each other's Phase 1 evaluations and engage in structured peer discussion.

Input: Each expert sees ALL other experts' evaluations (but not their own · they already have it)

Output structure:

- `RESPONSES`: Per-expert targeted responses ("Re: [Expert]: [Response]")
- `UPDATED_ASSESSMENT`: Revised view after considering peers
- `BLIND_SPOTS`: Issues not yet adequately addressed
- `CONFIDENCE`: Updated confidence score

Design rationale: Expert-to-expert discussion forces cross-examination. An expert who was overly confident must address challenges from peers. An expert who missed a factor must acknowledge it. This mimics the dynamics of a real expert committee.

5.5 Phase 4: Final Assessment

After the full discussion, each expert gives their final assessment incorporating all information from Phases 1-3.

Output structure:

- `FINAL_ASSESSMENT`: 2-3 sentence conclusive evaluation
- `RECOMMENDATION`: Concrete, actionable advice
- `RISK_WARNING`: 1-2 key risks the user should monitor
- `CONFIDENCE`: Final confidence score

5.6 Report Synthesis

The system aggregates all Phase 4 outputs into a structured consultation report:

1. **Expert Panel Summary**: Each expert's final assessment

2 **Key Insights**: Unique insights from each expert's domain
3 **Blind Spots**: Issues identified during discussion
4 **Recommendations**: Actionable items from all experts
5 **Risk Warnings**: Consensus and divergent risk assessments
6 **Confidence Shift**: How expert confidence changed from Phase 1 to Phase 4

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6. Information Retrieval and Grounding

6.1 RAG Pipeline

The system retrieves current information to ground expert analysis:

1. **YouTube Transcript Retrieval**: Searches for videos related to the topic, extracts full transcripts via `YouTubeTranscriptApi`. Longer videos (indicating deeper analysis) are prioritized.
2. **News Search**: Tavily API provides recent news articles for supplementary context.
3. **RAFT-style Document Mixing**: Documents are split into "oracle" (high-quality, relevant) and " distractor" (potentially irrelevant) categories, then shuffled. Experts must evaluate document relevance as part of their analysis.

6.2 Information Partitioning

To create genuine information asymmetry among experts:

```
Total documents
    ··· 40% · Shared pool (all experts see these)
    ··· 60% · Private pool
                ··· Agent 1: 2 exclusive documents
                ··· Agent 2: 2 exclusive documents
                ··· Agent 3: 2 exclusive documents
                ··· Agent 4: 2 exclusive documents
                ··· Agent 5: 2 exclusive documents
```

This ensures experts have different information bases, making their independent evaluations genuinely diverse rather than superficially diverse.

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7. Technical Implementation

7.1 System Requirements

- **Python 3.11+**
- **LLM Backends**: DeepSeek V3 (primary), with optional OpenAI GPT-4o, Anthropic Claude, Google Gemini
- **APIs**: Tavily (news search), YouTube Data API (transcript retrieval)
- **Frontend**: Streamlit with Plotly for visualization
- **Execution**: ThreadPoolExecutor for parallel agent execution

7.2 Core Components

| Component | File | Responsibility | |-----------|------|----------------| | `ConsultAgent` | `consult_agent.py` | Individual expert agent with 4-phase methods | | `ConsultationPipeline` | `consultation.py` | Pipeline

orchestration and report synthesis | | `build_clone_prompt` | `knowledge_loader.py` | Level 2 knowledge profile injection | | `Expert Panels` | `expert_panels.py` | Domain-specific expert configurations | | `NewsRetriever` | `retriever.py` | RAG pipeline (YouTube + news) | | `partition_documents` | `info_partition.py` | Information asymmetry creation | | Streamlit App | `app_consult.py` | User interface and session management |

7.3 Multi-Model Distribution

Agents are distributed across available LLM backends in round-robin fashion:

```
for i, expert in enumerate(panel):
    backend = backends[i % len(backends)] if backends else None
    agents.append(ConsultAgent(**expert, backend=backend))
```

This provides:

- **Diversity of reasoning**: Different LLMs have different reasoning patterns
- **Cost optimization**: Mix expensive and affordable models
- **Rate limit management**: Distribute API calls across providers
- **Resilience**: If one backend fails, others continue

7.4 Parallel Execution

All phases use `ThreadPoolExecutor` with `max_workers=len(agents)` for full parallelism:

```
with ThreadPoolExecutor(max_workers=len(self.agents)) as pool:
    futures = {pool.submit(_run, a): a for a in self.agents}
    for future in as_completed(futures):
        agent = futures[future]
        results[agent.name] = future.result()
```

Typical wall-clock time for a full consultation: 2-4 minutes (vs. 10-20 minutes sequential).

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8. Use Cases

8.1 Cryptocurrency Investment Analysis

Scenario: A user believes "Bitcoin will reach $150,000 this year due to ETF inflows and the halving effect."

Expert responses (illustrative):

- ··:·"MVRV Z-Score at 2.8 suggests room for upside, but LTH supply distribution hasn't reached typical top-cycle patterns. The $150K target requires sustained supply contraction."
- ··:·"ETF inflows are demand-side positive, but $150K implies global liquidity expansion that isn't guaranteed. If inflation rebounds, rate cut expectations · a key BTC driver · could collapse.
- ··:·"Historical max drawdowns of 77-93% mean you should ask: 'Can I hold through a 50% drop on the way to $150K?' Position sizing matters more than price targets."
- ··:·"The ETF approval was a milestone, but ongoing SEC litigation against exchanges creates regulatory uncertainty. A surprise enforcement action could trigger a 20%+ correction."

Value delivered: The user's thesis isn't wrong, but they learn about supply-side constraints, macro dependencies, position sizing risks, and regulatory tail risks they hadn't considered.

8.2 Sports Prediction Analysis

Scenario: A user believes "The Lakers will definitely make the playoffs because they have LeBron."

Expert responses (illustrative):

- ··:·"Current Net Rating of +1.2 implies ~62% playoff probability. Not 'definitely' · nearly 4 in 10 simulations have them missing."
- ··:·"AD's 27% career absence rate is the key variable. When AD is out, the team's Net Rating drops by 10 points · from playoff team to lottery team."
- ···:·"The market has them at -150 to make playoffs (implied 60%). If you think 'definitely' (>90%), the market disagrees by 30 percentage points."
- ···:·"Having a star doesn't guarantee playoffs · see 2019 Lakers (LeBron, 33-49 record). System fit and health matter as much as talent."

Value delivered: The user's confidence is calibrated from "definitely" (~95%) to a more realistic 60-65%, with specific variables to monitor.

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9. Evaluation Framework

9.1 Proposed Metrics

We propose evaluating the framework across three dimensions:

1. Opinion Calibration

- Measure user confidence before and after consultation
- For verifiable predictions: compare user's post-consultation confidence to actual outcomes using Brier Score
- Target: post-consultation Brier Score < pre-consultation Brier Score

2. Blind Spot Discovery

- Count unique risk factors / considerations identified by experts that the user hadn't mentioned
- Measure user acknowledgment of new factors in Phase 2 interactions
- Target: · 3 new considerations per consultation

3. Expert Diversity

- Measure inter-expert agreement (lower = more diverse perspectives)
- Track confidence spread across experts
- Measure stance coverage: all 4 stance types represented in the output
- Target: no two experts give identical assessments

9.2 Comparison Baselines

| Baseline | Description | |----------|-------------| | Single LLM | Same question to one LLM instance | | Single LLM + " Devil's Advocate" prompt | One LLM asked to challenge the user | | Level 1 Multi-Agent | 5 agents with persona descriptions only (no knowledge profiles) | | **Our System (Level 2)** | 5 agents with knowledge profiles + 4-phase pipeline |

9.3 Experimental Design (Proposed)

1. **Dataset**: 50 verifiable prediction questions (25 crypto, 25 sports) with known outcomes
2. **Participants**: Simulated users with pre-set opinions (varying accuracy)
3. **Metrics**: Pre/post Brier Score, blind spot count, expert diversity index

4. **Hypothesis**: Level 2 cloning + multi-phase pipeline produces better calibration than all baselines

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10. Limitations and Future Work

10.1 Current Limitations

1. **Knowledge Profile Staleness**: Few-shot examples in knowledge profiles are static. They don't update with market conditions or new events.
2. **No Ground Truth Feedback Loop**: The system doesn't track whether its advice led to better outcomes. A feedback mechanism would enable Level 3 cloning via fine-tuning.
3. **Language**: Current implementation is Chinese-language. Internationalization would expand the user base.
4. **Expert Selection**: Panels are pre-defined per domain. Ideally, the system would dynamically select experts based on the specific topic.
5. **Cost**: A full 4-phase consultation with 5 experts requires ~40 LLM calls (5 agents × 4 phases × 2 including meta). At current API prices, this is $0.05-0.50 per consultation depending on model.

10.2 Future Directions

Level 3 Cloning: Fine-tune LoRA adapters on curated expert corpora (e.g., all Arthur Hayes blog posts, all Willy Woo Twitter threads) for deeper behavioral fidelity.

Dynamic Expert Selection: Use a classifier to analyze the user's topic and automatically compose the optimal expert panel from a larger pool.

Persistent Expert Memory: Each expert maintains memory across consultations · remembering their past predictions, tracking their accuracy, and adjusting confidence accordingly. This connects to the existing `ReputationTracker` from the prediction market system.

Cognitive Bias Detection: Automatically detect confirmation bias (user only engages with agreeing experts), anchoring (user opinion doesn't change despite strong counter-evidence), and authority bias (user only trusts highest-confidence expert).

Multi-Turn Consultation: Allow users to return with updated opinions after real-world developments, creating a longitudinal consultation record.

Aggregation Mechanisms: Integrate the LMSR, BTS, and reputation-weighted aggregation from the original prediction market system to produce a mathematically grounded "consensus score" alongside the qualitative report.

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11. Conclusion

We have presented a multi-agent AI consultation framework that transforms the single-LLM advice paradigm into a structured expert panel experience. Through Level 2 AI Cloning (knowledge profile injection with few-shot examples), a 4-phase consultation pipeline (evaluate · interact · discuss · finalize), and designed diversity (varied expert stances and information asymmetry), the system produces richer, more balanced, and more actionable advice than any single AI interaction.

The framework is domain-agnostic · while we demonstrate cryptocurrency and sports implementations, the architecture supports any domain where expert consultation adds value: medical second opinions, legal strategy, career decisions, policy analysis, and more.

By making high-quality multi-expert consultation accessible to anyone, we aim to democratize the kind of structured analytical thinking that was previously available only to those with access to expensive human expert networks.

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Appendix A: Knowledge Profile Example (Abbreviated)

Expert: ······ (Risk Management Consultant) **Archetype**: Nassim Taleb's tail-risk thinking + Kelly Criterion

Core Framework:

1. Volatility analysis (historical vol, implied vol, ATR)
2. Drawdown analysis (MDD, recovery time)
3. Position sizing (Kelly Criterion, VaR/CVaR)
4. Correlation risk (dynamic correlation, tail correlation)
5. Liquidity risk (spread, depth, slippage)
6. Extreme scenarios (black swans, counterparty risk)

Few-shot Example (abbreviated):

> **User**: "I plan to allocate 50% of my portfolio to Bitcoin"
> **Expert Response**: "Let me quantify what this means with risk data. Bitcoin's annualized volatility is 60-80% (vs S&P 500 at 15-18%). A 50% allocation creates portfolio volatility of 30-40%. Historical drawdowns: -93% (2011), -86% (2014), -84% (2018), -77% (2022). At 50% allocation, a -77% BTC drawdown means -38.5% total portfolio loss. Kelly Criterion suggests optimal allocation of ~30% (half-Kelly for parameter uncertainty). Recommendation: max 20-30%, build position over 3-6 months, set portfolio-level max drawdown tolerance."

Typical Phrases:

- "Let me show you what this means in numbers"
- "If you're wrong, what's the worst case?"
- "Position sizing matters 10x more than direction"

Appendix B: System Prompt Composition

```
Total system prompt: ~3,000-3,500 characters per expert
Composition:
··· Layer 1 (Identity):        ~150 chars  (5%)
··· Layer 2 (Knowledge Profile): ~2,500 chars (80%)
·   ··· Archetype:              ~100 chars
·   ··· Framework:              ~500 chars
·   ··· Few-shot Examples (×3): ~1,500 chars
·   ··· Typical Phrases:        ~400 chars
··· Layer 3 (Behavioral Rules): ~300 chars  (10%)
··· Layer 4 (RAG Context):      Variable    (5%+)
```

Appendix C: API Cost Estimation

Phase	LLM Calls	Avg Tokens/Call	Est. Cost (DeepSeek V3)
Phase 1 (Evaluate)	5	~3,000 in + ~800 out	$0.005
Phase 2 (Chat, avg 2 turns × 2 experts)	4	~3,500 in + ~500 out	$0.004
Phase 3 (Discuss)	5	~4,000 in + ~800 out	$0.006
Phase 4 (Final)	5	~4,500 in + ~500 out	$0.006
RAG Retrieval	1	·	$0.001 (Tavily)
Total	**~20**		**~$0.02**

At DeepSeek V3 pricing ($0.27/M input, $1.10/M output), a full consultation costs approximately $0.02 · orders of magnitude cheaper than human expert consultation ($200-500/hour).

This whitepaper describes the Collective AI Deliberation framework as implemented in the open-source prediction-market-debater project.